COMPUTER NETWORKS – PRACTICAL JOURNAL

Measuring Throughput and Latency using ping and iperf3

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Aim

To measure the latency (delay) of a network using the ping command and the throughput (data rate) using the iperf3 tool, for both TCP and UDP traffic, in both directions.

Requirements

Windows PC with Command Prompt, iperf3 (Windows 64-bit build, folder iperf3.1.4_64 in C:\), and a network connection. For the two-PC version: two PCs in the same network (via switch, router or Wi-Fi) with the same iperf3 folder copied on both.

Basic Terms

Term	Meaning
Latency	Time taken by a packet to travel to the destination (and back, as round-trip time, RTT). Measured in milliseconds with ping.
Throughput	The actual amount of data transferred per second (Mbit/s, Gbit/s). Measured with iperf3.
Jitter	Variation in the delay of packets. Reported by iperf3 in UDP mode.
Packet loss	Percentage of packets that do not reach the destination. Shown by ping and by iperf3 UDP.
iperf3	A client-server tool: one side runs as the server (-s) and the other as the client (-c). Default port 5201, default test time 10 seconds.

Setup used in this journal

The commands below were run on a single computer with IP address 192.168.1.8, where both the iperf3 server and the client were running. Below each step, a box shows how the same step is done on two different PCs. In the two-PC boxes, PC-A is the server PC and PC-B is the client PC, and SERVER-IP stands for the IP address of PC-A.

PART A: Latency using ping

Step 1: Open Command Prompt (Win + R, type cmd, press Enter) and find the IP address of the PC with ipconfig (IPv4 Address = 192.168.1.8 here).

```
C:\Users\lachh> ipconfig
```

▶ **If you use two different PCs**

```
PC-A> ipconfig
PC-B> ipconfig
```

Difference: Run ipconfig on both PCs and note the IPv4 address of PC-A (the server). Both PCs must be in the same network (same subnet).

Step 2: Test the PC's own network software with the loopback address. Option -n 10 sends 10 packets.

```
C:\Users\lachh> ping 127.0.0.1 -n 10
```

Output:

```
C:\Users\lachh>ping 127.0.0.1 -n 10

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

▶ **If you use two different PCs**

```
PC-B> ping 127.0.0.1 -n 10
```

Difference: No difference. The loopback address always refers to the PC itself, so it never tests the network.

Step 3: Ping the PC's own IP address. Replies in under 1 ms, TTL=128 (Windows default) and 0% loss are seen.

```
C:\Users\lachh> ping 192.168.1.8 -n 10
```

Output:

```
C:\Users\lachh>ping 192.168.1.8 -n 10

Pinging 192.168.1.8 with 32 bytes of data:
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.8:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

▶ **If you use two different PCs**

```
PC-B> ping SERVER-IP -n 10
```

Difference: Ping the other PC's address instead of your own. The packets now travel through the real network (cable/switch/Wi-Fi), so the round-trip time is the real LAN latency and is usually more than 0 ms (typically a few ms on Wi-Fi). It also confirms that the two PCs can reach each other before running iperf3.

Step 4: Ping an internet host to measure internet latency. Here google.com answered in about 7 ms, TTL=113.

```
C:\Users\lachh> ping google.com -n 10
```

Output:

```
C:\Users\lachh>ping google.com -n 10

Pinging google.com [142.250.146.101] with 32 bytes of data:
Reply from 142.250.146.101: bytes=32 time=7ms TTL=113
Reply from 142.250.146.101: bytes=32 time=7ms TTL=113
Reply from 142.250.146.101: bytes=32 time=7ms TTL=113
Reply from 142.250.146.101: bytes=32 time=6ms TTL=113
Reply from 142.250.146.101: bytes=32 time=8ms TTL=113
Reply from 142.250.146.101: bytes=32 time=8ms TTL=113
Reply from 142.250.146.101: bytes=32 time=7ms TTL=113
Reply from 142.250.146.101: bytes=32 time=7ms TTL=113
Reply from 142.250.146.101: bytes=32 time=7ms TTL=113
Reply from 142.250.146.101: bytes=32 time=7ms TTL=113

Ping statistics for 142.250.146.101:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 8ms, Average = 7ms
```

▶ **If you use two different PCs**

```
PC-A> ping google.com -n 10
PC-B> ping google.com -n 10
```

Difference: No difference in the command. Each PC measures its own path to the internet, so the times can differ slightly.

PART B: Throughput using iperf3

Step 5: Go to the iperf3 folder. Typing iperf3 directly from C:\Users\lachh gave “iperf3 is not recognized”, because the program is not in the PATH. Move to the folder that contains iperf3.exe.

```
C:\Users\lachh> iperf3 -c 192.168.1.8
'iperf3' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\lachh> cd ../..
C:\> cd iperf3.1.4_64
C:\iperf3.1.4_64>
```

▶ **If you use two different PCs**

```
PC-A> cd C:\iperf3.1.4_64
PC-B> cd C:\iperf3.1.4_64
```

Difference: The same iperf3 folder (same version) must be present on both PCs, since both ends of the test use iperf3.exe.

Step 6: Start the iperf3 server in the first Command Prompt window. It waits on port 5201. (Typing iperf3 -s 192.168.1.8 gave “unable to start listener … Address already in use”, because a server was already listening on port 5201; only one server can use a port, and -s does not take an IP address — use -B to bind to one.)

```
C:\iperf3.1.4_64> iperf3 -s
```

Output:

```
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
```

▶ **If you use two different PCs**

```
PC-A> iperf3 -s
(allow port 5201 in the firewall, run once as Administrator:)
netsh advfirewall firewall add rule name="iperf3" dir=in action=allow
protocol=TCP localport=5201
netsh advfirewall firewall add rule name="iperf3" dir=in action=allow
protocol=UDP localport=5201
```

Difference: The server runs only on PC-A, and PC-B runs the client. The Windows Firewall on PC-A must allow port 5201 (TCP and UDP), otherwise the client on PC-B shows “Connection timed out”.

Step 7: Open a second Command Prompt window for the client, go to the same folder and run the default TCP test (client sends data to the server for 10 seconds).

```
C:\iperf3.1.4_64> iperf3 -c 192.168.1.8
```

Output:

```
Connecting to host 192.168.1.8, port 5201
[  4] local 192.168.1.8 port 62753 connected to 192.168.1.8 port 5201
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-1.01   sec   280 MBytes  2.34 Gbits/sec
[  4]   1.01-2.01   sec   225 MBytes  1.88 Gbits/sec
[  4]   2.01-3.01   sec   190 MBytes  1.60 Gbits/sec
[  4]   3.01-4.01   sec   312 MBytes  2.61 Gbits/sec
[  4]   4.01-5.01   sec   242 MBytes  2.03 Gbits/sec
[  4]   5.01-6.00   sec   237 MBytes  2.00 Gbits/sec
[  4]   6.00-7.01   sec   183 MBytes  1.52 Gbits/sec
[  4]   7.01-8.00   sec   178 MBytes  1.50 Gbits/sec
[  4]   8.00-9.01   sec   133 MBytes  1.10 Gbits/sec
[  4]   9.01-10.01  sec   209 MBytes  1.76 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-10.01  sec  2.14 GBytes  1.84 Gbits/sec    sender
[  4]   0.00-10.01  sec  2.14 GBytes  1.83 Gbits/sec    receiver

iperf Done.
```

The summary lines give the result: 2.14 GBytes transferred in 10 seconds, an average throughput of about 1.84 Gbit/s. On the server window, the same transfer is shown as received (1.83 Gbit/s).

▶ **If you use two different PCs**

```
PC-B> iperf3 -c SERVER-IP
```

Difference: Same command, but with the server PC’s IP address. The data really crosses the network, so the throughput is limited by the link: about 940 Mbit/s is typical for a gigabit Ethernet link, about 90–100 Mbit/s for Fast Ethernet, and Wi-Fi gives a lower and more variable value. The “local” address in the output will be PC-B’s IP, not the same address as the server.

Step 8: Reverse mode: the server sends and the client receives, so the other direction is measured. The option for this in iperf3 is -R (the option -r that was typed first is from iperf2 and gives “unknown option”).

```
C:\iperf3.1.4_64> iperf3 -c 192.168.1.8 -r -t 10
iperf3: unknown option -- r

C:\iperf3.1.4_64> iperf3 -c 192.168.1.8 -R -t 10
```

Output:

```
Connecting to host 192.168.1.8, port 5201
Reverse mode, remote host 192.168.1.8 is sending
[  4] local 192.168.1.8 port 62756 connected to 192.168.1.8 port 5201
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-1.00   sec   899 MBytes  7.54 Gbits/sec
[  4]   1.00-2.00   sec   734 MBytes  6.16 Gbits/sec
[  4]   2.00-3.01   sec   778 MBytes  6.45 Gbits/sec
[  4]   3.01-4.01   sec   522 MBytes  4.38 Gbits/sec
[  4]   4.01-5.01   sec   407 MBytes  3.42 Gbits/sec
```

```
[  4]   5.01-6.00   sec   597 MBytes  5.06 Gbits/sec
[  4]   6.00-7.01   sec   175 MBytes  1.46 Gbits/sec
[  4]   7.01-8.01   sec   245 MBytes  2.05 Gbits/sec
[  4]   8.01-9.00   sec   189 MBytes  1.60 Gbits/sec
[  4]   9.00-10.00  sec   224 MBytes  1.87 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-10.00  sec  4.66 GBytes  4.00 Gbits/sec    sender
[  4]   0.00-10.00  sec  4.66 GBytes  4.00 Gbits/sec    receiver

iperf Done.
```

Option -t 10 sets the test time to 10 seconds (the default). The result is 4.66 GBytes at an average of 4.00 Gbit/s.

▶ **If you use two different PCs**

```
PC-B> iperf3 -c SERVER-IP -R -t 10
```

Difference: The command is the same. With -R, PC-A sends and PC-B receives, so you measure the download speed of PC-B (the opposite direction of the previous test). On a real link, the two directions can give different results, for example on Wi-Fi or an asymmetric link.

Step 9: UDP test: option -u switches to UDP. By default, iperf3 sends UDP at only 1 Mbit/s, and reports jitter and datagram loss.

```
C:\iperf3.1.4_64> iperf3 -c 192.168.1.8 -u -t 10
```

Output:

```
Connecting to host 192.168.1.8, port 5201
[  4] local 192.168.1.8 port 60607 connected to 192.168.1.8 port 5201
[ ID] Interval           Transfer     Bandwidth       Total Datagrams
[  4]   0.00-1.01   sec   120 KBytes   978 Kbits/sec  15
[  4]   1.01-2.01   sec   128 KBytes  1.04 Mbits/sec  16
   ... (similar lines for 2-9 seconds) ...
[  4]   9.00-10.00  sec   128 KBytes  1.05 Mbits/sec  16
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Jitter    Lost/Total Datagrams
[  4]   0.00-10.00  sec  1.24 MBytes  1.04 Mbits/sec  0.084 ms  0/159 (0%)
[  4] Sent 159 datagrams

iperf Done.
```

Result: 159 datagrams sent, 0 lost (0%), jitter 0.084 ms.

▶ **If you use two different PCs**

```
PC-B> iperf3 -c SERVER-IP -u -t 10
PC-B> iperf3 -c SERVER-IP -u -b 100M -t 10    (optional)
```

Difference: The first command is the same. On two PCs, option -b 100M can be added to send at 100 Mbit/s instead of 1 Mbit/s, which shows the real packet loss and jitter of the link when it is loaded.

Step 10: UDP in reverse mode (server sends UDP, client receives).

```
C:\iperf3.1.4_64> iperf3 -c 192.168.1.8 -u -R -t 10
```

Output:

```
Connecting to host 192.168.1.8, port 5201
Reverse mode, remote host 192.168.1.8 is sending
[  4] local 192.168.1.8 port 62916 connected to 192.168.1.8 port 5201
[ ID] Interval           Transfer     Bandwidth       Jitter    Lost/Total Datagrams
[  4]   0.00-1.01   sec   120 KBytes   976 Kbits/sec  0.091 ms  0/15 (0%)
[  4]   1.01-2.01   sec   128 KBytes  1.05 Mbits/sec  0.060 ms  0/16 (0%)
   ... (similar lines for 2-9 seconds) ...
[  4]   9.00-10.00  sec   128 KBytes  1.05 Mbits/sec  0.046 ms  0/16 (0%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Jitter    Lost/Total Datagrams
[  4]   0.00-10.00  sec  1.24 MBytes  1.04 Mbits/sec  0.046 ms  0/159 (0%)
[  4] Sent 159 datagrams
```

```
iperf Done.
```

▶ **If you use two different PCs**

```
PC-B> iperf3 -c SERVER-IP -u -R -t 10
```

Difference: Same command; the server PC-A now sends the UDP stream and PC-B measures the loss and jitter while receiving.

Step 11: Check the latency again with ping after the tests. Option -n takes the number of packets, so ping -n 192.168.1.8 -n 10 gives “IP address must be specified”; give the address once and then -n 10.

```
C:\iperf3.1.4_64> ping -n 192.168.1.8 -n 10
IP address must be specified.

C:\iperf3.1.4_64> ping 192.168.1.8 -n 10
```

Output:

```
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128   (10 replies)

Ping statistics for 192.168.1.8:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

▶ **If you use two different PCs**

```
PC-B> ping SERVER-IP -n 10
```

Difference: Compare this ping with the one from Step 3. You can also run ping while iperf3 is running to see how latency increases when the link is loaded.

Step 12: Stop the iperf3 server in the first window by pressing Ctrl + C. The server window had shown each test it received (TCP forward, TCP reverse, UDP forward and UDP reverse).

▶ **If you use two different PCs**

```
PC-A> Ctrl + C   (stop the server)
```

Difference: Only the server PC (PC-A) needs to be stopped. Remove the firewall rule afterwards if it was added: netsh advfirewall firewall delete rule name="iperf3".

Observations

Test	Command	Result
Ping loopback	ping 127.0.0.1 -n 10	10/10 replies, 0% loss, <1 ms, TTL=128
Ping own IP	ping 192.168.1.8 -n 10	10/10 replies, 0% loss, <1 ms, TTL=128
Ping internet	ping google.com -n 10	0% loss, 6–8 ms (average 7 ms), TTL=113
TCP (client sends)	iperf3 -c 192.168.1.8	2.14 GBytes in 10 s, 1.84 Gbit/s
TCP reverse	iperf3 -c 192.168.1.8 -R -t 10	4.66 GBytes in 10 s, 4.00 Gbit/s
UDP	iperf3 -c 192.168.1.8 -u -t 10	1.04 Mbit/s, jitter 0.084 ms, loss 0/159 (0%)
UDP reverse	iperf3 -c 192.168.1.8 -u -R -t 10	1.04 Mbit/s, jitter 0.046 ms, loss 0/159 (0%)

Note on single-PC results: Because the client and server run on the same computer (192.168.1.8), the packets never leave the PC. The very high speeds (Gbit/s) and latency under 1 ms therefore show the speed of the PC itself, not of a network link, and the TCP rate changes from second to second because both programs compete for the same CPU. To measure a real network, repeat the steps on two PCs as given in the orange boxes.

Result

✓ EXPERIMENT COMPLETED SUCCESSFULLY

Latency was measured with ping (loopback, own IP and an internet host), and throughput was measured with iperf3 for TCP and UDP in both directions. The tests showed 0% packet loss and the throughput values listed in the observation table.

VIVA QUESTIONS AND ANSWERS

A. Latency, Throughput and Basic Concepts

Q1. What is latency?

Ans: Latency is the time a packet takes to go from the source to the destination. With ping we usually measure the round-trip time (RTT), which is the time for the request to reach the destination and the reply to come back. It is measured in milliseconds; in our test, google.com showed about 7 ms.

Q2. What is throughput?

Ans: Throughput is the actual rate at which data is successfully transferred between two devices, measured in bits per second (Mbit/s, Gbit/s). It is what iperf3 reports as Bandwidth, for example 1.84 Gbit/s in the TCP test.

Q3. What is the difference between bandwidth and throughput?

Ans: Bandwidth is the maximum capacity of the link (for example 1 Gbit/s), while throughput is the speed really achieved, which is usually lower because of protocol overheads, errors, congestion, CPU limits and Wi-Fi conditions.

Q4. What is the difference between latency and throughput?

Ans: Latency is how long it takes for data to arrive (delay), and throughput is how much data arrives per second (rate). A satellite link can have high throughput but high latency; a dial-up link has low latency but low throughput.

Q5. What are the components of latency?

Ans: Propagation delay (signal travelling through the medium), transmission delay (time to put the bits on the link), queuing delay (waiting in router buffers) and processing delay (time taken by devices to handle the packet).

Q6. What is jitter?

Ans: Jitter is the variation in the delay between packets. It matters for voice and video. iperf3 reports it in UDP tests; in our test the jitter was 0.084 ms (forward) and 0.046 ms (reverse).

Q7. What is packet loss?

Ans: The percentage of packets that never reach the destination. ping shows it as “Lost = 0 (0% loss)” and iperf3 UDP shows it as “Lost/Total Datagrams 0/159”. Loss reduces throughput and quality.

Q8. What is the bandwidth-delay product?

Ans: Bandwidth multiplied by the round-trip delay. It is the amount of data that can be “in flight” on the link, and TCP needs a window at least this large to fill the link.

Q9. Which factors affect throughput?

Ans: Link speed and duplex, Wi-Fi signal and interference, TCP window size, latency, packet loss, CPU load of the PCs, network congestion, MTU and the protocol overhead.

Q10. Why do we measure latency and throughput?

Ans: To check the performance of a network, find bottlenecks, compare links (Ethernet vs Wi-Fi), size the network for applications such as video calls, and to troubleshoot slowness.

B. ping

Q11. What is ping and which protocol does it use?

Ans: ping is a tool that checks whether a host is reachable and measures RTT. It uses ICMP echo request and echo reply messages.

Q12. What does the option -n 10 do in ping?

Ans: On Windows, -n sets the number of echo requests to send (here 10). On Linux the same option is -c. Without it, Windows sends 4 requests; -t makes it ping continuously.

Q13. Why did “ping -n 192.168.1.8 -n 10” give “IP address must be specified”?

Ans: The option -n expects a number after it, so 192.168.1.8 was taken as the count and no target remained. The correct form is ping 192.168.1.8 -n 10.

Q14. What is the loopback address 127.0.0.1?

Ans: It is a special address that always points to the same computer. The traffic never goes out to the network, so the test only checks the PC’s own TCP/IP software. It always gave <1 ms in our test.

Q15. Why does ping to our own IP 192.168.1.8 also give <1 ms?

Ans: The operating system recognises its own address and delivers the packet internally, without sending it to the network card. So it is as fast as the loopback and not a real network measurement.

Q16. What does TTL mean in the ping reply?

Ans: Time To Live is a counter in the IP header which every router decreases by 1; if it reaches 0 the packet is discarded. It stops packets from looping. The value in the reply tells about the replying OS and the distance.

Q17. Why is the TTL 128 for our own IP and 113 for google.com?

Ans: Windows starts packets with TTL 128. Our own PC replied directly, so it stayed 128. For google.com the reply crossed several routers and each one reduced the TTL, giving 113. The exact hop count cannot be read from the TTL alone, as the starting value of the remote host is not known.

Q18. What do “time<1ms” and “time=7ms” mean?

Ans: The round-trip time was measured as less than 1 millisecond (too small to measure on Windows), and exactly 7 ms for the internet host. The minimum, maximum and average RTT are shown in the statistics.

Q19. What does the ping statistics summary show?

Ans: Packets sent, received and lost (with percentage), and the minimum, maximum and average round-trip times. For google.com: 10 sent, 10 received, 0% loss, 6 ms to 8 ms, average 7 ms.

Q20. Can ping measure throughput?

Ans: No. ping measures latency and loss with small packets; it does not measure the data rate. iperf3 is used for throughput.

Q21. What if ping shows “Request timed out”?

Ans: No reply came within the timeout: the host may be down, unreachable, blocked by a firewall (ICMP blocked), or there is a network problem.

C. iperf3 Basics and Options

Q22. What is iperf3?

Ans: iperf3 is a command-line tool for measuring the maximum achievable network throughput between two hosts. One host runs as server and the other as client, and the tool sends test data for a set time and reports the speed, and for UDP the jitter and loss.

Q23. What is the difference between iperf and iperf3?

Ans: iperf3 is a complete rewrite of iperf (iperf2) with a simpler code and some different options, and the two are not compatible with each other. For example, -r (tradeoff) in iperf2 does not exist in iperf3, which uses -R and --bidir.

Q24. Why did “iperf3 -c 192.168.1.8 -r” give “unknown option -- r”?

Ans: -r is an iperf2 option. In iperf3 reverse mode (server sends, client receives) is -R in capital letters. The usage list printed by iperf3 shows -R, --reverse.

Q25. Why was “iperf3 is not recognized” shown at first?

Ans: The Command Prompt was in C:\Users\lachh and iperf3.exe is not installed in a folder listed in the PATH. We changed to the iperf3 folder (cd iperf3.1.4_64) or the folder can be added to the PATH.

Q26. What does iperf3 -s do?

Ans: It starts iperf3 in server mode. The server listens (default port 5201) and waits for a client to connect, then receives or sends the test data and shows the result.

Q27. What does iperf3 -c <ip> do?

Ans: It starts iperf3 in client mode and connects to the server with the given IP address, then generates the traffic for the test (10 seconds by default) and prints the interval and final results.

Q28. Why did "iperf3 -s 192.168.1.8" give "Address already in use"?

Ans: Another iperf3 server was already listening on port 5201 in the other window. Only one program can listen on a port at a time. Close the old server or use another port with -p. (To bind a server to a specific IP, the option is -B.)

Q29. What is the default port of iperf3?

Ans: 5201, for both TCP and UDP. It can be changed with -p on both server and client. This port must be allowed through the firewall.

Q30. What does the option -t do?

Ans: It sets the duration of the test in seconds; the default is 10 seconds. -t 10 was used in our tests.

Q31. What does the option -R do?

Ans: Reverse mode: the server sends the data and the client receives it. It tests the opposite direction without changing the server and client roles. The output says "Reverse mode, remote host … is sending".

Q32. What does the option -u do?

Ans: It makes iperf3 use UDP instead of TCP. By default, it sends 1 Mbit/s and shows jitter and lost datagrams.

Q33. What does the option -b do?

Ans: It sets the target bandwidth. For UDP the default is 1 Mbit/s, so -b 100M sends 100 Mbit/s; for TCP it is unlimited by default. It is used to load a link in UDP tests.

Q34. What does the option -P do?

Ans: It runs several parallel streams (for example -P 4). It is useful to fill a fast link when a single TCP stream cannot.

Q35. What is the use of -i, -f, -w, -B, -p, -n, -J and -V?

Ans: -i sets the report interval in seconds, -f sets the format (K, M, G), -w sets the TCP window/buffer size, -B binds to an interface address, -p changes the port, -n sends a fixed number of bytes instead of a time, -J gives JSON output and -V gives more detailed output.

Q36. What do the columns Interval, Transfer and Bandwidth mean?

Ans: Interval is the time slice of the test, Transfer is the amount of data sent in that interval and Bandwidth is the rate in that interval (in Mbit/s or Gbit/s). The last lines give the whole-test average for sender and receiver.

Q37. What is the difference between the sender and receiver lines?

Ans: Sender is the speed measured at the sending side and receiver is what the other side actually received. They are almost equal when there is no loss, as in our TCP test (1.84 and 1.83 Gbit/s).

Q38. What do the numbers 62753 and 5201 in the output mean?

Ans: 5201 is the server port. 62753 is the temporary (ephemeral) port chosen by the client for this connection. Each new test uses a new client port.

Q39. Can iperf3 measure latency?

Ans: Not directly. ping is used for latency. iperf3 gives jitter and loss in UDP mode, which are related quality measures.

D. TCP vs UDP Results

Q40. Why do we test both TCP and UDP?

Ans: TCP shows the maximum reliable throughput with flow and congestion control. UDP shows how a link handles a fixed-rate stream and gives jitter and packet loss, which matter for voice and video.

Q41. What is the difference between TCP and UDP?

Ans: TCP is connection-oriented, reliable, retransmits lost data and controls the rate. UDP is connectionless, has no retransmission or congestion control, and is faster and lighter but unreliable.

Q42. Why did the UDP test show only about 1 Mbit/s?

Ans: iperf3 sends UDP at 1 Mbit/s by default. The rate is not the limit of the network; to test a faster rate, use -b (for example -b 100M).

Q43. What does "0/159 (0%)" mean in the UDP output?

Ans: Out of 159 datagrams sent, 0 were lost, i.e. 0% loss. The total in 10 seconds follows from the 1 Mbit/s rate and the 8 KB datagram length.

Q44. Why did the TCP throughput change in every second (1.10 to 2.61 Gbit/s)?

Ans: TCP adjusts its rate with congestion control, and both iperf3 processes were sharing the same CPU and memory on one PC. The scheduling of the two programs changes second to second, so the rate varies.

Q45. Why was the throughput so high (Gbit/s) in our test?

Ans: The client and the server were on the same PC, so data moved through memory without using a network cable or Wi-Fi. The speed shows how fast the PC can copy data, not the network speed.

Q46. Why was the reverse test (4.00 Gbit/s) faster than the forward test (1.84 Gbit/s)?

Ans: On a single PC there is no real network, so the difference comes from how Windows scheduled the sending and receiving processes and buffers. On two PCs, a difference would reflect real differences between the upload and download direction of the link.

Q47. What speed can be expected on a real network?

Ans: On gigabit Ethernet, TCP usually gives about 940 Mbit/s because of header overhead. Fast Ethernet gives about 94 Mbit/s. Wi-Fi gives much lower, variable numbers depending on signal and standard.

Q48. What is TCP window size and how does it affect throughput?

Ans: The window is the amount of data that can be sent before an acknowledgement is needed. A small window on a high-latency link limits the throughput to window ÷ RTT, so larger windows (-w) or parallel streams (-P) can increase it.

Q49. What is MTU?

Ans: Maximum Transmission Unit, the largest packet size a link can carry (1500 bytes for Ethernet). It determines the TCP segment size and so affects efficiency and throughput.

Q50. What is the effect of Wi-Fi interference on the results?

Ans: It causes retransmissions, higher latency and jitter, and lower throughput that varies with time. Wired Ethernet is more stable.

E. Two-PC Setup and Scenarios

Q51. What changes when we use two PCs instead of one?

Ans: The server is run on PC-A (iperf3 -s) and the client on PC-B with the IP address of PC-A (iperf3 -c SERVER-IP). The data now crosses the real network, so throughput, latency, jitter and loss show the actual link performance.

Q52. Which IP address is given in the client command on two PCs?

Ans: The IP address of the server PC (PC-A), found with ipconfig on PC-A. It must be reachable from PC-B, which can be checked with ping first.

Q53. What if the client shows “Connection timed out” or “Connection refused”?

Ans: The server is not running, the IP address or port is wrong, the PCs are in different networks, or the firewall of the server PC blocks port 5201. Start the server and allow the port in Windows Firewall.

Q54. How do we allow iperf3 in the Windows Firewall?

Ans: Create an inbound rule for TCP and UDP port 5201, e.g. netsh advfirewall firewall add rule name="iperf3" dir=in action=allow protocol=TCP localport=5201 (and the same for UDP), run in Command Prompt as Administrator.

Q55. How can we test both directions on two PCs?

Ans: Run the client on PC-B normally to test the upload from PC-B to PC-A, and add -R to test the download from PC-A to PC-B. The server stays on PC-A.

Q56. Should both PCs be in the same subnet?

Ans: For a simple test, yes, for example 192.168.1.x with the mask 255.255.255.0. If they are in different subnets, a router with proper routing between them is required.

Q57. Which is better for throughput testing: wired or Wi-Fi?

Ans: Wired Ethernet gives stable, repeatable results close to the link speed. Wi-Fi results change with distance, interference and the number of users, so repeat the test several times.

Q58. How can we see latency under load?

Ans: Run a long iperf3 test (for example -t 30) and at the same time run ping to the server. If the RTT rises, the link is congested (queuing delay increases).

Q59. Why should a test be repeated several times?

Ans: The results vary with network load and CPU usage. The average of several runs is more reliable than one run.

Q60. What is the effect of increasing the test time?

Ans: A longer test (-t 30 or more) gives a more stable average, and shows changes in throughput over time, but takes longer.

Q61. What is the use of Ctrl + C in this experiment?

Ans: It stops the running server (or any running command). The iperf3 server keeps listening for new clients until it is stopped.